ECE2132 Data Structures & Algorithms Lab — Start here

Built from the 9 manual files you uploaded (Exp 1–8, 10–12; AY 2026-27). Every program in this pack was compiled with g++ in strict mode and run on the manual's own sample inputs. The output boxes are copied from those real runs.

C++ from zero → Pointers (1–2) → Arrays (3–4) → Matrices (5–6) → Sorting (7–8) → Stack (9) → Queue (10) → Linked lists (11–12) → Exam day

What the lab exam asks of you

You will be given one experiment (possibly with a small change, like the manual's ASSIGNMENT lines). To score, you need four things:

1. **Write the record part**: aim, short algorithm, time/space complexity, result.
2. **Type the program** from memory, compile it, and fix your own typos from the compiler's messages.
3. **Run it** on a sample input and show the right output.
4. **Answer viva questions**: why this line, what happens if…, what is the complexity.

Each experiment chapter gives you those four, in that order.

How to use each chapter

> **THE ONE RULE**
>
> **Guess before you read.** Each chapter opens with a question box. Answer it in your head or on paper *before* reading on, even if you're unsure. A wrong guess still helps: the right answer sticks better after you've tried. Then type the program yourself, without looking. Re-reading feels like studying but doesn't make you able to type it from scratch.

- **Code boxes** (light grey) contain the corrected program. Type it yourself; don't copy-paste.
- **Screen boxes** (black) show what the program printed in a real run, with the typed input included.
- **Red boxes** show real compiler errors from the manual's broken versions, so you learn to read them.
- **Check** questions close each chapter. Answers are at the end of that chapter.
- **"If the examiner changes the question"** sections hold the manual's ASSIGNMENT variants. Each one is tested.

Manual errors found (and fixed in this pack)

The manual's listings are images, and many experiments print the code twice (a PROGRAM listing and an IMPLEMENTATION screenshot) that **don't match**. Everything below was checked by compiling and running it.

Exp	Problem in the manual	Effect	Fix in this pack
2	RESULT paragraph copied from Exp 1	Wrong result statement	Own result written
3	No check on position or free space	Position 0 or 50 corrupts memory	Position 1…n+1 checked
4	Every option asks "Enter value:" and reads a number it never uses (listing only; the screenshot is right)	With the manual's own sample input, the program waits forever for a value	Prompt removed; deleted element printed
4	Algorithm step 5 says "insertion menu"	Text slip	—
5	Algorithm step 7 is a stray deletion line; explanation table rows are shifted by one	Text slips	Correct table in chapter
6	Fig. 3 caption "A[i][j] to T[i][j]"	Should be T[j][i]	—
7, 8	`int arr[n];` with `n` read at run time	Not standard C++. Visual Studio refuses it; g++ only allows it as an extension	`int arr[100];` + range check
8	Aim and explanation promise Bubble, Selection, Insertion, `copyArray()` and `clock()` timing, but the code has only Quick Sort	Half the experiment is missing	Full comparison program added
8	Heading "Characteristics of Insertion Sort" over quick sort points	Text slip	—
10	Figure captions say "…Operation on **Stack**"	Copied from a stack experiment	—
10	Screenshot copy: `enqueue(value);` outside its `if`, choice 3 prints "Program terminated", bare `else`	Doesn't compile	Listing version used, braces fixed
10	Array named `queue`	Clashes with `std::queue` if `<bits/stdc++.h>` is included	Renamed `queueArr`
11	Screenshot copy: traverse never moves `temp` (infinite loop), insert never links the node, menu text corrupted	Broken	Listing version used (it is correct)
12	Listing: `insert(intvalue)`, menu strings without closing quotes, `delete temp;` outside the block where `temp` exists, `main` cut off	Doesn't compile	Fixed and completed

> **EXPERIMENT 9 IS MISSING**
>
> No "Exp 9" file was uploaded. Exp 10's figure captions read "Enqueue Operation on Stack", so Exp 9 is almost certainly **Stack using arrays**. Chapter 8 of this pack covers it, written in the same style as the Queue experiment. It is **not** from your manual; check with a classmate whether Exp 9 really is Stack.

The compiler on your laptop

To practise at home you need a C++ compiler. One is now installed on this laptop: **g++ 16.2 (WinLibs MinGW)**. Chapter 1 shows how to use it. In the lab you'll use whichever editor is there (VS Code, Code::Blocks or Dev-C++). The programs in this pack run the same on all three.